Quantification: 542_z18
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	12
interior lacuna count	11
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 11)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	3688.27	3684	1441.75	px^2
major_axis_length_px	lacuna shape	132.275	128.431	35.9804	px
minor_axis_length_px	lacuna shape	35.8241	35.4482	8.8535	px
aspect_ratio	lacuna shape	3.6839	3.6773	0.6253	unitless
eccentricity	lacuna shape	0.9593	0.9623	0.0137	unitless
solidity	lacuna shape	0.8643	0.893	0.0736	unitless
roots_count	headline	12.5455	10	6.2026	count
ring_length_r30_px	headline	518.273	557	171.563	px
ring_length_r60_px	reported	1209.73	1284	380.128	px
owned_length_px	ownership-dependent	3093.8	3212.03	1109.94	px
edge_count	ownership-dependent	142.364	157	56.4451	count
mean_edge_length_px	ownership-dependent	22.7155	21.7526	3.7529	px
perimeter_px	normalised	349.688	337.706	124.859	px
ring_area_r30_px2	normalised	12006.2	12118	2958.11	px^2
ring_area_r60_px2	normalised	28554.3	29514	6304.69	px^2
in_frame_fraction_r30	normalised	0.9834	1	0.0361	unitless
in_frame_fraction_r60	normalised	0.9555	1	0.0828	unitless
ring_density_r30	normalised	0.0422048	0.042351	0.00800895	px^-1
ring_density_r60	normalised	0.0414477	0.0435398	0.00817982	px^-1
roots_per_100px_perimeter	normalised	3.5845	3.2405	1.0197	per 100 px
ring_attached_length_r30_px	network v2	482.636	505	179.677	px
ring_attached_length_r60_px	network v2	1050.73	1115	399.938	px
ring_length_w_r30_px	network v2	583.743	624.831	197.985	px
ring_length_w_r60_px	network v2	1369.07	1445.74	436.232	px
sholl_crossings_r10	network v2	13.0909	14	4.8053	count
sholl_crossings_r20	network v2	16	17	5.9833	count
sholl_crossings_r30	network v2	16.7273	17	5.2553	count
loop_count	loops, report only	12.8182	14	6.7501	count
loop_edge_share	loops, report only	0.3538	0.3171	0.1112	share
ring_width_mean_r30_px	width	7.5863	7.4935	0.5112	px

Per field
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measure	value	units
analysed_area_px2	1.0074e+06	px^2
total_skeleton_length_px	39799	px
canalicular_length_density_per_px	0.0395065	px^-1
junction_count	922	count
junction_density_per_px2	0.00091522	px^-2
lacuna_count	12	count

lacunae_per_px2	1.191e-05	px^-2
skeleton_component_count	227	count
mean_component_length_px	175.326	px
median_component_length_px	22	px
field_length_density_w_per_px	0.0446835	px^-1
field_density_without_flagged_per_px	0.039462	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	372	px
unattached_component_count	22	count
canalicular_length_density_attached_per_px	0.0395065	px^-1
field_length_density_with_unattached_per_px	0.0398758	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	171	count
loop_edge_share	0.3475	share
edge_count	1994	count

Canalicular width, the new measure

measure	value	units
width_mean_px	8.2591	px
width_median_px	8.2462	px
width_p10_px	4.4721	px
width_p90_px	10.7703	px
width_pixels_used	38209	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	602	56.2113	21.2037	2.651	0.9261	0.6645	-0.3574	16.2608	914.042	True
2	5924	181.043	48.2611	3.7513	0.9638	0.675	-0.0924	129.871	638.627	False
3	4438	160.681	35.4482	4.5328	0.9754	0.8989	-0.2181	173.351	382.514	False
4	4342	127.796	43.8553	2.914	0.9393	0.893	-0.1983	210.3	874.108	False
5	5254	148.687	46.2758	3.2131	0.9503	0.9227	-0.0688	369.252	79.7008	False
6	3405	124.602	35.4524	3.5146	0.9587	0.8958	-0.0028	532.24	401.315	False
7	4133	149.961	37.557	3.9929	0.9681	0.861	-0.1428	589.071	924.049	False
8	2572	119.294	30.2433	3.9445	0.9673	0.8307	-0.0493	570.032	785.133	False
9	3684	159.337	32.9692	4.8329	0.9784	0.8054	-0.098	739.452	247.536	False
10	3421	128.431	34.9252	3.6773	0.9623	0.8774	-0.2145	813.822	732.524	False
11	2890	110.803	33.408	3.3167	0.9535	0.9299	0.0034	944.816	407.904	False
12	508	44.3947	15.6697	2.8331	0.9356	0.917	-0.2083	990.155	937.699	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	4	241	606	669.144	38	17.6091	6.9616
2	26	754	1600	2857.8	144	19.8458	7.4838
3	10	557	1289	3415.16	157	21.7526	7.997
4	20	626	1412	4956.59	241	20.5667	7.1625
5	11	506	1180	2537.55	134	18.937	8.1993

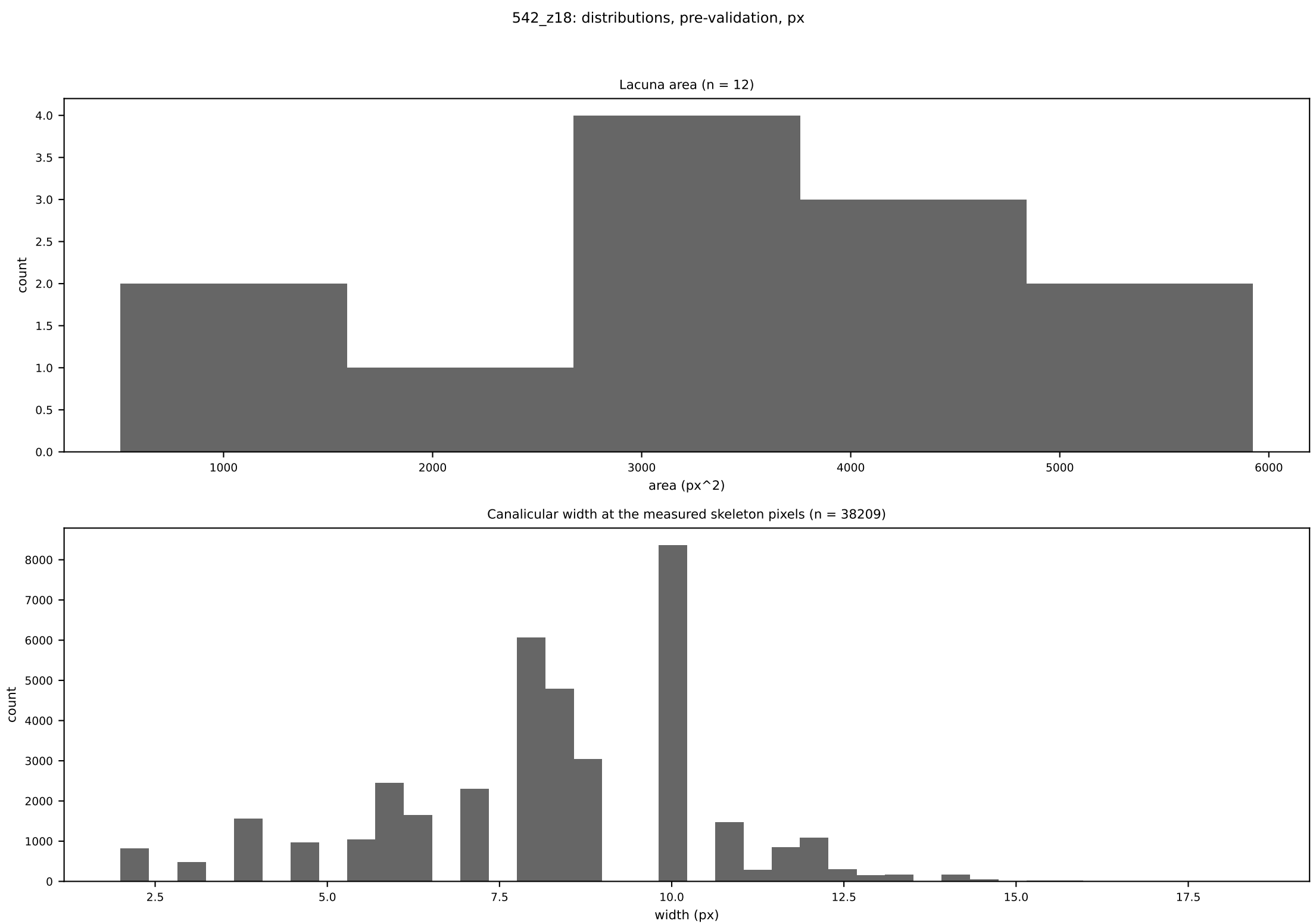
6	10	474	1248	4074.13	175	23.2808	8.1557
7	15	664	1619	3099.08	161	19.249	8.1319
8	8	497	1139	3512.78	141	24.9133	7.4935
9	16	608	1284	3212.03	160	20.0752	7.0786
10	10	579	1479	3825.43	166	23.0447	7.2211
11	7	288	708	1424.47	46	30.9668	7.832
12	5	148	349	1116.76	41	27.2381	6.6935

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	142.811	4575	11641	0.673	0.6207	0.0526776	0.0520574	2.8009
2	625.725	16860	35430	0.9542	0.8861	0.0447212	0.0451595	4.1552
3	377.019	13152	31926	1	1	0.042351	0.0403746	2.6524
4	337.706	12118	29514	1	1	0.0516587	0.0478417	5.9223
5	371.806	13199	31872	1	0.9956	0.0383362	0.0370231	2.9585
6	308.593	11379	28385	1	1	0.0416557	0.0439669	3.2405
7	382.877	12760	30245	1	1	0.0520376	0.0535295	3.9177
8	306.492	10697	26160	1	1	0.0464616	0.0435398	2.6102
9	437.262	14880	35351	1	1	0.0408602	0.0363215	3.6591
10	326.836	11837	29271	1	1	0.0489144	0.0505278	3.0596
11	263.48	9968	22850	0.9798	0.8783	0.0288925	0.0309847	2.6567
12	108.77	5218	13093	0.8837	0.75	0.0283634	0.0266555	4.5969

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	234	478	274.551	682.872	5	7	6
2	734	1304	857.11	1827.96	20	24	23
3	505	1109	624.831	1445.74	13	17	19
4	608	1214	705.357	1619.55	16	20	20
5	403	981	563.889	1332.53	8	15	17
6	445	1100	534.375	1429.45	11	14	17
7	642	1553	748.5	1817.3	18	23	20
8	497	1115	553.919	1280.21	14	16	16
9	578	1117	699.156	1464.31	16	17	17
10	553	1435	655.701	1669.84	16	18	21
11	211	374	322.664	793.097	7	9	8
12	133	256	155.669	379.764	5	3	6


542_z18: distributions, pre-validation, px
Lacuna area (n = 12)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 38209)
count
width (px)